

PE3101P: Decision and Social Choice

Optional Exercises 1 — Solutions

Tomi Francis

1.
 - (a) **Both.** A very likely winner at 14:1 is a good bet on the evidence, and it won.
 - (b) **Only in the fact-relative sense.** A small chance of dying at any moment does not justify spending money she expects to need. But she died before the contract expired, so the savings would have gone unused.
 - (c) **Neither.** Turning down heavily subsidised insurance against an outcome that would bankrupt him is a bad decision on the evidence, and he fell ill.
 - (d) **Both.** He knew the opposing commander was very cautious, and the alternative was to give up the city he had been ordered to defend. And it worked.
 - (e) **Neither.** He died. And it is foolish to accept a $1/20,000$ chance of death to save a few seconds.
 - (f) **Only in the evidence-relative sense.** Given his condition and a real chance that someone was infectious, refusing was right on the evidence. In fact nobody had the virus, so going would have been fine.
 - (g) **Only in the fact-relative sense.** Giving up a cheap and highly effective measure against a significant risk, for a very small gain elsewhere, is a bad decision on the evidence. But no war occurred, so the measure would have bought nothing.

2.
 - (a) **Where to put your money.**

Acts: buy stocks; buy bonds.

States: there is a stock market crash this year; there is no stock market crash this year.

Outcomes: the investor is wiped out; the investor makes a very healthy return; the investor makes a small return.

	Crash	No crash
Buy stocks	Wiped out	Very healthy return
Buy bonds	Small return	Small return

- (b) **How to get to the airport.**

Acts: take a taxi; take the airport train.

States: there is heavy traffic on the expressway; there is no heavy traffic.

Outcomes: you miss your flight, having spent \$30; you catch your flight, having spent \$30; you catch your flight, having spent \$12.

	Heavy traffic	No heavy traffic
Take a taxi	Miss it, spent \$30	Catch it, spent \$30
Take the airport train	Catch it, spent \$12	Catch it, spent \$12

The train is unaffected by road traffic, so it gets you there in both states. You pay for the taxi whether or not it gets you there in time.

(c) **Which crop to plant.**

Acts: plant a frost-resistant crop; plant a drought-resistant crop.

States: an early frost; a normal season; a drought.

Outcomes: a normal harvest; the crop dies.

	Early frost	Normal season	Drought
Frost-resistant crop	A normal harvest	A normal harvest	The crop dies
Drought-resistant crop	The crop dies	A normal harvest	A normal harvest

Each crop survives the condition it resists and a normal season, and dies in the other extreme.

3. (a) **Not jointly exhaustive.** The two dice can show the same number, and no column covers that.
 - (b) **Not mutually exclusive.** It can rain while the traffic is heavy, and then two columns apply at once and disagree. The states are exhaustive: “neither” covers the rest.
 - (c) **Nothing is wrong with it.** There is a delay or there is not, never both and never neither; how you travel cannot affect whether the line is delayed; and each act together with each state fixes both when you arrive and what you pay.
 - (d) **The states are not independent of the acts.** Locking the bicycle affects whether it is stolen. As the table stands, leaving it unlocked strongly statewise dominates locking it.
 - (e) **The states do not settle the outcome.** “Buy a ticket” together with “the draw is fair” leaves it open whether you win. To repair the table, make the states about which numbers are drawn.
 - (f) **The states are not independent of the acts.** Training hard does not determine whether you are injured, but it raises the probability, which is enough. As the table stands, training hard weakly statewise dominates training lightly, ignoring that it makes the bad state more likely.
4. (a) Suppose A strongly statewise dominates B .

In every state, A 's outcome is better than B 's. [definition of strong dominance]

So in every state, A 's outcome is at least as good as B 's. [better implies at least as good]

And in at least one state, A 's outcome is better than B 's. [there is at least one state]

So A weakly statewise dominates B . [definition of weak dominance]

- (b) (i) A_1 strongly dominates A_2 . A_1 weakly dominates A_3 (better in S_1 , equal in S_2). A_3 weakly dominates A_2 (equal in S_1 , better in S_2). A_2 dominates nothing.
- (ii) **Nothing dominates anything.** A_1 is much better in S_1 and S_2 but worse in S_3 , which is enough to rule out dominance. Dominance takes no account of how probable the states are.
- (iii) A_4 strongly dominates A_1 , A_2 and A_3 . A_1 weakly dominates A_3 , and so does A_2 . Neither of A_1 and A_2 dominates the other: their rows are identical, and weak dominance requires being better in at least one state.

5. (a) $\emptyset$ and X are complements, so:

$$\begin{aligned}\text{Cr}(\emptyset) &= \text{Cr}(\neg X) \\ &= 1 - \text{Cr}(X) && \text{[worked example]} \\ &= 1 - 1 = 0 && \text{[Normalisation]}\end{aligned}$$

- (b) Suppose for contradiction that $\text{Cr}(\sigma) > 1$.

$$\begin{aligned}\text{Cr}(\neg\sigma) &= 1 - \text{Cr}(\sigma) && \text{[worked example]} \\ &< 0 && \text{[since } \text{Cr}(\sigma) > 1\text{]}\end{aligned}$$

But $\text{Cr}(\neg\sigma) \geq 0$ by Non-negativity. Contradiction; so $\text{Cr}(\sigma) \leq 1$.

- (c) σ and τ are each disjoint from ρ , so $\sigma \cup \tau$ is disjoint from ρ . Hence:

$$\begin{aligned}\text{Cr}(\sigma \cup \tau \cup \rho) &= \text{Cr}((\sigma \cup \tau) \cup \rho) \\ &= \text{Cr}(\sigma \cup \tau) + \text{Cr}(\rho) && \text{[Additivity]} \\ &= \text{Cr}(\sigma) + \text{Cr}(\tau) + \text{Cr}(\rho) && \text{[Additivity, } \sigma \text{ and } \tau \text{ disjoint]}\end{aligned}$$

- (d)

$$\begin{aligned}\text{Cr}(\tau) &= \text{Cr}(\sigma \cup (\tau \setminus \sigma)) && \text{[given, since } \sigma \subseteq \tau\text{]} \\ &= \text{Cr}(\sigma) + \text{Cr}(\tau \setminus \sigma) && \text{[Additivity, the two being disjoint]} \\ &\geq \text{Cr}(\sigma) && \text{[Non-negativity]}\end{aligned}$$

- (e)

$$\begin{aligned}\text{Cr}(\sigma) &= \text{Cr}((\sigma \cap \tau) \cup (\sigma \cap \neg\tau)) && \text{[given]} \\ &= \text{Cr}(\sigma \cap \tau) + \text{Cr}(\sigma \cap \neg\tau) && \text{[Additivity, the two being disjoint]}\end{aligned}$$

- (f) First:

$$\begin{aligned}\text{Cr}(\sigma \cup \tau) &= \text{Cr}(\sigma \cup (\tau \setminus \sigma)) && \text{[given]} \\ &= \text{Cr}(\sigma) + \text{Cr}(\tau \setminus \sigma) && \text{[Additivity, the two being disjoint]}\end{aligned}$$

Next:

$$\begin{aligned}\text{Cr}(\tau) &= \text{Cr}((\tau \setminus \sigma) \cup (\sigma \cap \tau)) && \text{[given]} \\ &= \text{Cr}(\tau \setminus \sigma) + \text{Cr}(\sigma \cap \tau) && \text{[Additivity, the two being disjoint]}\end{aligned}$$

so $\text{Cr}(\tau \setminus \sigma) = \text{Cr}(\tau) - \text{Cr}(\sigma \cap \tau)$. Substituting into the first result:

$$\text{Cr}(\sigma \cup \tau) = \text{Cr}(\sigma) + \text{Cr}(\tau) - \text{Cr}(\sigma \cap \tau).$$